

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.*

1. Prove or disprove the following.

- a. A constant function is not one-to-one if the domain has more than one element.

True. Let $f: A \rightarrow B$ be a constant function and $|A| > 1$. Since $|A| > 1$ it has at least two elements, say a_1 and a_2 . f is constant so then $f(a_1) = f(a_2) = b$ for some $b \in B$. Thus f is not one-to-one.

Note that if the domain has only one element then this statement is false. Let $A = \{a\}$ and $f: A \rightarrow B$ and $f(a) = b$ for all $a \in A$. If $f(a_1) = f(a_2)$ then $a_1 = a_2$ because A only has one element. So f is constant and also one-to-one. *In general constant functions are not 1-1, i.e., the statement "constant functions are 1-1" is false.* $\square$

- b. The identity function is one-to-one.

True. Let $f: A \rightarrow A$ be the identity function. If $f(a_1) = f(a_2)$ then $f(a_1) = a_1 = a_2 = f(a_2)$ because $f(a) = a$ for any $a \in A$. Thus f is one-to-one. $\square$

- c. The identity function is onto.

True. Let $f: A \rightarrow A$ be the identity function and $a \in A$. $f(a) = a$ so then f is onto. $\square$

2. Let $g: A \rightarrow B$ and $f: B \rightarrow C$. Prove that if $f \circ g: A \rightarrow C$ is one-to-one and g is onto then f is also one-to-one.

Proof. Let $b_1, b_2 \in B$ such that $b_1 \neq b_2$. Because g is onto there is a $a_1, a_2 \in A$ such that $g(a_1) = b_1$ and $g(a_2) = b_2$. So then $f(b_1) \neq f(b_2)$ since $f(g(a_1)) \neq f(g(a_2))$. $\square$

3. Suppose that $f: A \rightarrow B$ is a one-to-one function. Prove or disprove that if $f^{-1}: f(B) \rightarrow A$ is a function then f^{-1} is also one-to-one.

True. Recall: It is shown in the notes that for a function g , the inverse of g is a function if and only if g is 1-1. Since f^{-1} is a function and $(f^{-1})^{-1} = f$ is also a function, it follows that f^{-1} is 1-1. $\square$

Alternatively,

Proof. Suppose to the contrary that f^{-1} is not 1-1. So then there is $b_1, b_2 \in B$ such that $b_1 \neq b_2$ and $f^{-1}(b_1) = f^{-1}(b_2) = a$ for some $a \in A$. Thus $f(f^{-1}(b_1)) = b_1 \neq b_2 = f(f^{-1}(b_2))$ which contradicts the assumption that f is a function. $\square$

4. Let $f: A \rightarrow B$ be a one-to-one function. Show that $|A| \leq |B|$.

Proof. Let $|A|$ be the number of items and $|B|$ be the number of boxes. By the pigeon hole theorem, if $|A| > |B|$ then f would map (or place) at least 2 items to the same box and thus not be 1-1. Thus f must be 1-1. $\square$

5. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Prove that $g \circ f : A \rightarrow C$ is a function.

Proof. f is a function implies that for any $a \in A$ there is a unique $b \in B$ such that $f(a) = b$. Also, g is a function implies that for any $b \in B$ there is a unique $c \in C$ such that $g(b) = c$. Thus for any $a \in A$ there is a unique $c \in C$ such that $g(f(a)) = g(b) = c$. So $g \circ f$ is a function. $\square$

6. (This problem was dropped) Let $f : A \rightarrow A$, $g : A \rightarrow A$, and $h : A \rightarrow A$; prove that $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. Let $a_0 \in A$. Since f, g, h are functions there are $a_1, a_2, a_3 \in A$ such that, $f(a_0) = a_1$, $g(a_1) = a_2$, and $h(a_2) = a_3$. $((h \circ g) \circ f)(a_0) = h(g(f(a_0))) = h(g(a_1)) = h(a_2) = a_3$. Also, $(h \circ (g \circ f))(a_0) = h((g \circ f)(a_0)) = h(g(f(a_0))) = h(g(a_1)) = h(a_2) = a_3$. Thus $(h \circ g) \circ f = h \circ (g \circ f)$. $\square$

7. Prove that if $f : A \rightarrow B$ is onto and $g : B \rightarrow C$ is onto, then $(g \circ f) : A \rightarrow C$ is onto.

Proof. Let $c \in C$. g is onto so then there exists a $b \in B$ such that $g(b) = c$. Since f is onto there also exists an $a \in A$ such that $f(a) = b$. So then $g(f(a)) = g(b) = c$; thus $g \circ f$ is onto. $\square$

8. Each of the following statements, $P(x, y)$, define a relation R in the natural numbers, $\mathbb{N}$, i.e., $R = (\mathbb{N}, \mathbb{N}, P(x, y))$. Show whether each of the following relations is reflexive.

a. $P(x, y) = \text{"}x \text{ is less than or equal to } y\text{"}$

Reflexive: Let $x \in \mathbb{N}$. $x = x \Rightarrow x \leq x$ so then R is reflexive. $\square$

b. $P(x, y) = \text{"}x \text{ divides } y\text{"}$

Reflexive: Let $x \in \mathbb{N}$. $x = 1 \cdot x$ so then $x|x$. Thus R is reflexive. $\square$

c. $P(x, y) = \text{"}x + y = 10\text{"}$

Not Reflexive: Consider the natural number 4; $4 + 4 \neq 10$. Thus R is not reflexive. $\square$

d. $P(x, y) = \text{"}x \text{ and } y \text{ are relatively prime."}$

Not Reflexive: Consider the natural number 8; $\gcd(8, 8) = 8 \neq 1$ since $8|8$. Thus R is not reflexive. $\square$